Remix VM Rehearsal Results

ECE 473 rain-count market · rehearsal of docs/deploy_sepolia.md · run on 12 November 2026, 23:02–23:32 UTC (6:02–6:32 pm New York)

Summary

- The full deploy guide (steps 1–28) ran end to end in the Remix VM. Every number the guide predicts matched to the wei: funding, quote, shares, prices before and after the buy, settlement, payout and sponsor withdrawal.
- Every safety check fired with the expected message: early report, early cancel, second redeem and a refund claimed by the wrong account.
- Three differences are in the current Remix interface, not in the contracts. Each has a simple fix, listed in the last section. All three would also affect the Sepolia run.
- This was the Remix VM (a practice chain in the browser with test accounts). Nothing was deployed to Sepolia and no real or test ETH was used.

Setup

Item	Value
Remix	Remix IDE 2.6.14 (app.remix.live), environment Remix VM, EVM version Osaka (default)
Compiler	0.8.26+commit.8a97fa7a, optimization on, 200 runs
Source files	contracts/remix/SeasonCountOracle_flat.sol and RainCountLMSR_flat.sol from github.com/qomer0507/ece-473 (main, commit 3cc2b7f)
Reporter (Account 1)	0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
Sponsor (Account 2)	0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2
Buyer (Account 3)	0x4B20993Bc481177ec7E8f571ceCaE8A9e22C02db
Oracle	0xd9145CCE52D386f254917e481eB44e9943F39138
Market (season 2025)	0xa131AD247055FD2e2aA8b156A11bdEc81b9eAD95
Cancel market (season 2028)	0x652c9ACcC53e765e1d96e2455E618dAaB79bA595
Times (main run)	T = 1794524536; seasonStart = tradingClose = 1794525436 (T+900); seasonEnd = 1794526036 (T+1500); reportDeadline = 1794527836 (T+3300)
Times (cancel run)	T2 = 1794525048; seasonStart = tradingClose = T2+300; seasonEnd = T2+420; reportDeadline = T2+600

The Remix VM used the real clock: `currentTime()` returned 1794524536 when the computer clock read 1794524541, so the guide's waiting times apply unchanged.

Main run: replay of the 2025 Trenton season (steps 1–23)

Step	Check	Guide expects	Observed	Result
1–3	Deploy oracle, openSeason(TTN, 2025, ...)	Succeeds	Both succeeded	**Match**
8	Deploy market with 0.018 ETH	Succeeds	Succeeded	**Match**
9	requiredFunding	17917594692280550	17917594692280550	**Match**
9	funding	18000000000000000	18000000000000000	**Match**
9	state / getEdges / bucketOf(5)	0 / 2,4,5,6,8 / 2	0 / 2,4,5,6,8 / 2	**Match**
9	prices (before buy)	six times 166666666666666666	six times 166666666666666666	**Match**

Step	Check	Guide expects	Observed	Result
11	quoteBasket([0,2,6,9,15,27])	1025582224601672	1025582224601672	**Match**
13	buyBasket with VALUE 0.00105 ETH	Succeeds	First UI try reverted: "Insufficient payment" (see Finding 1). Succeeded when VALUE was actually sent; gas 353,808	**Differs**
15	sharesOfAll(buyer)	0,2,6,9,15,27	0,2,6,9,15,27	**Match**
15	paid(buyer) and totalPaid	1025582224601672	1025582224601672 (both)	**Match**
15	prices (after buy)	150420933493721421, 153459637918370109, 159722444549102369, 164586717159761721, 174764191356159266, 197046075522885112	Identical, all six	**Match**
15	Contract balance	0.019025582 ETH	19025582224601672 wei	**Match**
17	reportCount before seasonEnd	Refused: "Season not over"	Refused: "Season not over"	**As expected**
18–19	reportCount(TTN, 2025, 5) after seasonEnd; hasValidReport	Succeeds; true	Succeeded; true	**Match**
20–21	settle; state / settledCount / winningBucket / reserved	1 / 5 / 2 / 600000000000000	1 / 5 / 2 / 600000000000000 (default gas was enough)	**Match**
22	redeem (Buyer)	0.0006 ETH paid out; reserved = 0	Succeeded; reserved went from 600000000000000 to 0	**Match**
22	Second redeem	Refused: "Nothing to redeem"	Refused: "Nothing to redeem"	**As expected**
23	sponsorWithdraw: SponsorWithdrawal amount	18425582224601672	18425582224601672	**Match**

Cancel run: nobody reports (steps 24–28)

Step	Check	Guide expects	Observed	Result
24–25	openSeason(TTN, 2028, ...); deploy second market with 0.018 ETH	Succeeds	Both succeeded	**Match**
26	Buyer buys the same hedge	Pays the quote	paid = 1025582224601672	**Match**
27	cancelAfterDeadline before the deadline	Refused: "Deadline not passed"	Refused: "Deadline not passed"	**As expected**
27	cancelAfterDeadline after the deadline	Succeeds; state = 2	With Remix's automatic gas: refused, "Send more gas" (see Finding 3). With gas limit 300000: succeeded, state = 2, gas 106,570 (guide's local figure: 106,570)	**Differs**
28	claimRefund (Buyer)	Full refund	Succeeded; paid(buyer) went from 1025582224601672 to 0	**Match**
28	claimRefund sent by the Sponsor by mistake	(not in guide)	Refused: "Nothing to refund"	**As expected**
28	sponsorWithdraw	Funding returned (0.018 ETH)	Succeeded; Sponsor balance rose from 99.963 to 99.981 ETH	**Match**

What the replay shows (for the slides)

- **Ordinary season (2025, 5 loss days):** the buyer paid 0.001026 ETH for the hedge and received 0.0006 ETH (6 shares in bucket 2). The sponsor kept 0.000426 ETH.
- **The market reacts to demand:** after one buy, the price of the wettest bucket rose from 16.7% to 19.7%, and the driest buckets fell to about 15%.
- **The sponsor's loss is capped:** the funding of 0.017918 ETH (b × ln 6 × unit) is the most the sponsor can lose, and the contract refuses to deploy with less.
- **Nobody reports, nobody loses:** when no report arrives by the deadline, the buyer gets back exactly what they paid and the sponsor gets back the funding.
- Per the deploy guide, a wet season such as 2009 or 2013 (12 loss days) would pay the same hedge 27 shares, or 0.0027 ETH. This case was not run in the rehearsal.

Differences from the guide, and fixes for the Sepolia run

#	What happened	Fix for the guide
1	The VALUE box at the top of the Deploy tab is only used for deployments in Remix 2.6.14. A buyBasket clicked with VALUE set there reverted with "Insufficient payment".	Each contract under **Deployed Contracts** has its own **Value** and **Gas limit** fields at the bottom of its box (below "Low level interaction"). Put the 0.00105 ETH there before clicking buyBasket. Tested: the UI buy then succeeded.
2	VALUE does not reliably go back to 0. The Deploy-tab Value stayed at 0.018 ETH after the second market deploy, and a contract's own Value stayed at 0.00105 ETH after the buy.	After every payable call, set the Value you used back to 0 yourself.
3	cancelAfterDeadline with Remix's automatic gas reverted with "Send more gas". The market requires at least ORACLE_GAS + 30000 gas left before it calls the oracle, and the automatic estimate falls just short. settle, which does the same check, worked with the automatic gas.	Before cancelAfterDeadline, set that contract's **Gas limit** to 300000 (already the fix in the guide's error table). Do the same for settle if it ever shows "Send more gas".

Notes

- While the Value field was being located, three transactions were sent from Remix's script terminal instead of the buttons, from the same accounts with the same arguments and amounts: the main buyBasket, the cancel market's buyBasket, and cancelAfterDeadline with claimRefund on the cancel market. Contract behaviour is the same either way.
- Gas: buyBasket used 353,808 in the Remix VM (guide's local figure 354,120). cancelAfterDeadline used 106,570 (identical). Other gas figures were not recorded.
- An extra test market (season 2029) was deployed only to confirm Fix 1. It is not part of the guide.
- The Remix VM state is lost if the page is reloaded. A VM backup demo has to be set up fresh, ideally with shorter season times.